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Enhancing Skin Regeneration during Expansion: A Multicenter Randomized Controlled Trial of Stromal Vascular Fraction and Fat Grafting

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Background: Skin regeneration by means of mechanical stretching is used in reconstructive surgery, but it is often limited by the skin's inherent growth capacity. This multicenter randomized controlled trial evaluated the efficacy of autologous stromal vascular fraction (SVF) transplantation and fat grafting in enhancing skin regeneration during tissue expansion.

Methods: Patients aged 18 to 60 years undergoing skin expansion were randomized to receive SVF transplantation or fat grafting or serve as controls. Participants were also categorized into well-regenerated and poorly regenerated subgroups based on skin texture assessments. Assessments occurred every 4 weeks over 12 weeks, with safety follow-up up to 2 years. The primary outcome was skin thickness change at 12 weeks; secondary outcomes included intermediate skin thickness and the expansion index.

Results: Seventy-two patients were enrolled; after 6 were lost to follow-up, 66 remained (23 in the control group, 21 in the adipose group, and 22 in the SVF group). At 12 weeks, the adipose and SVF groups showed significant increases in skin thickness compared with the control group ($P < 0.05$). In well-regenerated skin, treatments maintained thickness; in poorly regenerated skin, treatments increased and maintained thickness through 12 weeks. Both treatment groups had greater increases in expansion index at 12 weeks compared with controls ($P < 0.001$). No severe adverse events were observed during the 2-year follow-up.

Conclusions: Autologous SVF transplantation and fat grafting effectively promote skin regeneration during tissue expansion, maintaining skin thickness in well-regenerated skin and counteracting thinning in poorly regenerated skin. Adipose-derived treatments offer an effective strategy for enhancing skin regeneration in tissue expansion. *(Plast. Reconstr. Surg.* 157: 360, 2026.)

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Received for publication October 10, 2024; accepted July 23, 2025.
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This trial is registered under the name "Evaluation of the Effect of Autologous Fat and SVF Transplantation in Promoting Mechanical-stretch Induced In Vivo Skin Regeneration," ClinicalTrial.com registration no. NCT02336997 (https://clinicaltrials.gov/study/NCT02336997).

DOI: 10.1097/PRS.0000000000012347

Skin expansion is a common reconstructive technique used to repair deformities resulting from injury, disease, or congenital issues.[1] The process involves implanting an expander under the skin, and gradually inflating it to stimulate tissue growth through cellular proliferation and extracellular matrix (ECM) production.[2,3] However, the intrinsic limitations of skin growth often lead to complications, such as thinning and telangiectasia,[4,5] increasing the risk of treatment failure.[6,7]

Disclosure statements are at the end of this article, following the correspondence information.

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In clinical practice, skin expansion aims to produce sufficient skin for reconstructive procedures. Although treatments such as botulinum toxin type A and tanshinone IIA promote regeneration, they have associated risks.[8,9] Thus, innovative strategies are needed to improve skin growth without added complications. Autologous fat grafting has gained popularity across various medical specialties for tissue augmentation, and research shows that fat grafting enriched with adipose-derived stem cells significantly accelerates expansion.[10] Our prior clinical studies indicated that fat grafting effectively mitigates thinning and telangiectasia characteristic of papery skin during skin expansion.[11,12] The stromal vascular fraction (SVF) in adipose tissue, consisting of a heterogeneous population enriched with stem cells and progenitor cells, possessed regenerative properties.[13,14] Transplantation of SVF into expanded skin increases dermal and epidermal thickness, leading to improved texture and increased regeneration in expanded skin.[15,16]

Given the promising outcomes of preliminary studies, clinical trials focusing on SVF treatment and fat transplantation have been conducted in cases of skin expansion with deteriorated regeneration. Despite the demonstrated effectiveness of SVF treatment and fat transplantation in reviving poorly regenerated skin, further research is necessary to thoroughly evaluate and compare their effects. In this study, we propose a prospective, multicenter, randomized, clinical trial to assess the efficacy of autologous fat grafting and SVF transplantation in enhancing skin regeneration following expansion. Furthermore, we categorized participants into well-regenerated and poorly regenerated subgroups to determine treatment effects across different stages of regeneration. This trial aims to optimize the use of adipose-derived therapies in skin expansion, refining therapeutic strategies to improve clinical outcomes.

PATIENTS AND METHODS

Study Oversight

This multicenter, randomized, controlled trial, adhering to the Declaration of Helsinki, received ethical approval from the institutional ethics committees of all participating centers: the Ninth People's Hospital affiliated with Shanghai Jiao Tong University School of Medicine, the West China Hospital of Sichuan University, and The First Affiliated Hospital of Zhengzhou University. All treatments and follow-up visits were conducted at the departments of plastic and reconstructive surgery at each respective institution.

Study Design

Designed as a multicenter, parallel, single-blind trial, statistical power was projected to exceed 90% at a significance level of $\alpha = 0.05$ (2-sided), based on prior research.[15] A total of 60 participants were planned for enrollment, with 20 participants assigned to each group. Considering an expected 20% attrition rate, 72 participants were ultimately recruited, with 24 participants assigned to each group. Participants were recruited competitively across the 3 centers.

Inclusion and Exclusion Criteria

Eligible participants were adults aged 18 to 60 years, without sex bias, who had undergone expander implantation and required further expansion to meet the desired reconstruction. Exclusion criteria consisted of severe systemic conditions, infection at the expansion site or hair-bearing expansion locations, and recent nicotine use. Eligibility was assessed by 2 independent surgeons.

Randomization and Masking

All eligible patients provided written informed consent before randomization. Participants were then assigned randomly to the SVF transplantation, fat grafting, or control group in a 1:1:1 ratio using a computer-generated randomization schedule, ensuring allocation concealment across centers. Patients were further categorized into well-regenerated and poorly regenerated subgroups based on a clinical evaluation of the expanded skin texture.[4,5] Poorly regenerated skin was defined as skin exhibiting deterioration in expanded skin texture—such as thin or papery skin, telangiectasia, and dermal striae—that did not improve after a 2-week suspension of expansion. Well-regenerated skin did not show these characteristics. This assessment was conducted independently by 2 experienced independent senior plastic surgeons at each center. In cases of disagreement, a third expert provided a final evaluation to minimize subjectivity.

The recruitment team supervised the randomization process, whereas the conduct team administered the treatments at each center. Data collectors and technicians were masked to the group assignments. No substantial changes were made to the study design after the trial began.

Intervention Protocol

Autologous fat was harvested from the periumbilical abdomen or posterolateral thigh regions, using low-pressure liposuction.[17] The harvested fat was processed by washing and centrifugation to remove oil droplets, and the wetting solution and the purified adipose tissue was preserved. For transplantation, fat was grafted into precisely created tunnels between the dermis and the implant using a 20-gauge cannula. To ensure even distribution, each tunnel received 0.1 to 0.2 mL of injected fat at 0.5- to 1-cm intervals.[11]

SVF was isolated according to protocols established in our previous clinical study.[15] Each 20-mL fat sample was treated with collagenase (Shanghai Qiaoyuan Biological Pharmaceutical Co., Ltd., Shanghai, People's Republic of China) at 37°C for 1 hour, then filtered and centrifuged to isolate the red SVF cell pellet. After supernatant removal and saline washing, the cell concentration was adjusted to 1×10^6 cells/mL. To prevent accidental puncture of the expander, the inflated fluid was first withdrawn to loosen the expanded skin. Subsequently, the skin was gently pinched, and intradermal injections of the SVF suspension were administered—approximately 0.1 mL at 1-cm intervals (1×10^5 cells/cm^2) using a 30-gauge needle.

After 1 week of intervention, the expander was regularly inflated every 3 days throughout the study, with pressure monitored until it reached 100 mmHg. Inflation continued until sufficient tissue was acquired, or the maximum regenerative potential was achieved. A single intervention involving fat grafting and SVF transplantation was performed. The control group underwent the same assessment schedule as the other groups, but did not participate in the liposuction or injection procedures. Changes in the expanded skin were monitored without therapeutic intervention.

Outcome Assessments and Follow-Up

Assessments were conducted at baseline and subsequently at 4, 8, and 12 weeks after treatment at each center. Outcomes included changes in skin thickness at intermediate intervals and the expansion index (EI) at 12 weeks to assess the efficiency of the expansion process. Safety assessments were conducted throughout the follow-up period, extending up to at least 2 years.

Skin Thickness Measurement

Ultrasonography was performed on the expanded skin using B-mode ultrasonography (GE Healthcare, Chicago, IL) with an 8- to 12-MHz linear-array probe to measure skin thickness on average, at 8 intervals along the longitudinal axis of the expanded skin.[15,18] Measurements were performed by a trained technician who was blinded to group allocation.

EI Calculation

EI was defined as the ratio of the total inflated volume (in milliliters) to the designed volume of the expander (in milliliters), reflecting the efficiency of the expansion process. Inflated volumes were recorded and summed at each visit.

Safety Monitoring

Safety assessments were conducted throughout the 2-year follow-up period. All adverse events, both postinjection-related symptoms (eg, redness, swelling, pain, or infection) and long-term symptoms related to the presence of a subcutaneous protuberance, mass, or hyperplasia, were recorded.

Statistical Analysis

Statistical analysis was conducted using GraphPad Prism version 9.5 (GraphPad Software Inc., San Diego, CA). Normality was assessed using the D'Agostino and Pearson omnibus test. Continuous variables were expressed as mean ± SEM with 95% confidence intervals. Baseline data were tested using one-way analysis of variance. Subgroup analyses were prespecified and based on baseline skin regeneration status. Between-group differences at each visit were analyzed using 2-way repeated-measures analysis of variance, with factors for treatment group and time, followed by Sidak multiple comparison tests. Within-group differences at each follow-up visit compared with baseline were analyzed using the paired t test, followed by Dunnett multiple comparison tests. For subgroup analyses, interaction terms between treatment and subgroup were assessed to determine whether the treatment effects differed between the subgroups. Statistical significance was set at a value of $P < 0.05$ (*$P < 0.05$; **$P < 0.01$; and ***$P < 0.001$).

RESULTS

Participant Characteristics

From October of 2016 through June of 2022, 72 patients were screened and randomized into 3 groups: SVF, adipose, or control. After losing 6 patients to follow-up, the final cohort included

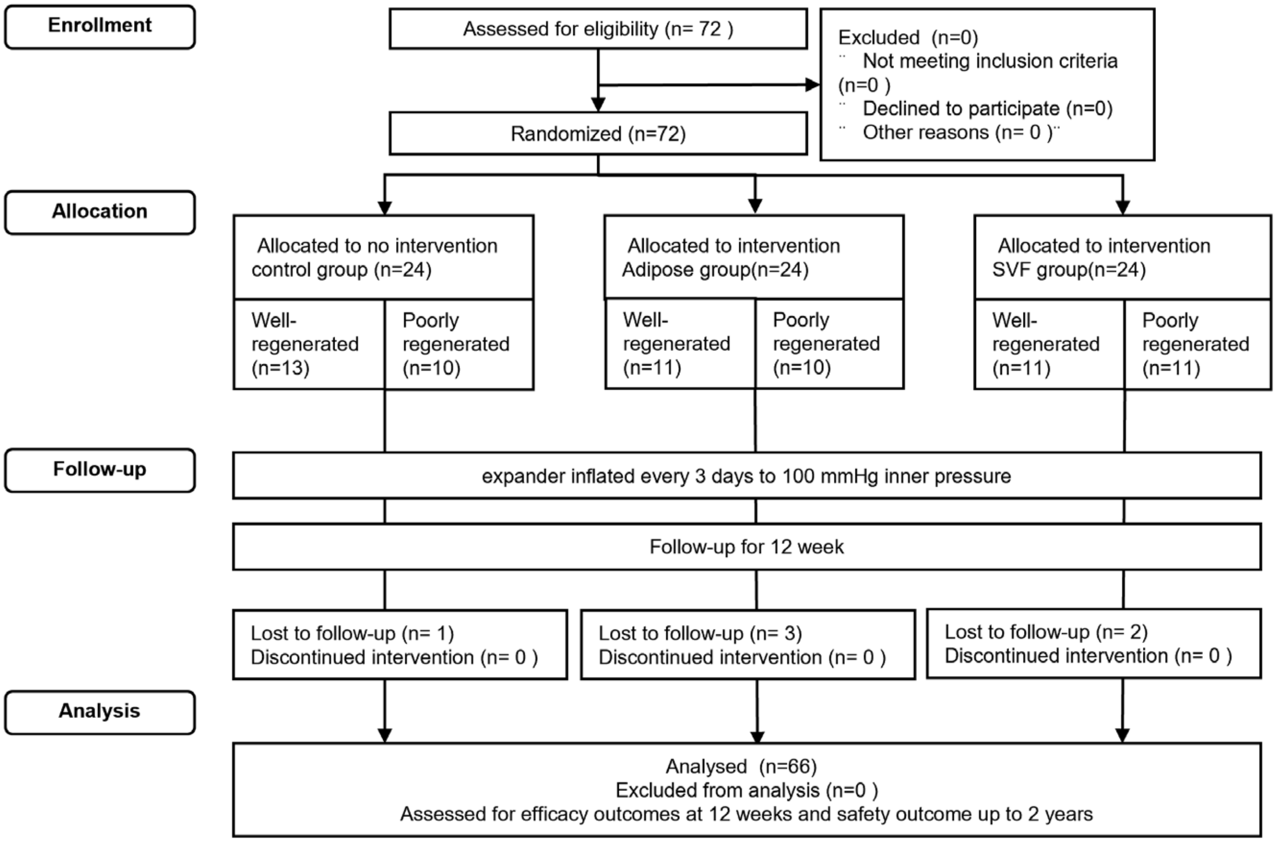

Fig. 1. Consolidated Standards of Reporting Trials flow diagram of the participant recruitment and study flow.

66 participants (23 in the control group, 21 in the adipose group, and 22 in the SVF group) (Fig. 1).

Patient demographics and baseline characteristics were consistent across all groups, as shown in Table 1. The most common indications for tissue expander placement were secondary deformities caused by burns, disease, or injury. There were no cases involving soft-tissue loss from cancer resection or patients who had undergone radiotherapy or chemotherapy, which might have affected tissue quality. The average ages were 30.9 ± 2.1 years in the control group, 28.2 ± 1.4 years in the adipose group, and 27.8 ± 1.4 years in the SVF group. Body mass index averages were 24.0 ± 0.9 kg/m^2 for the control group, 22.3 ± 0.8 kg/m^2 for the adipose group, and 21.4 ± 0.8 kg/m^2 for the SVF group. The EI was 1.58 ± 0.14 in the control group, 1.84 ± 0.17 in the adipose group, and 1.89 ± 0.20 in the SVF group. Baseline characteristics were homogeneous across groups, with no significant differences in age, body mass index, or initial EI.

Furthermore, based on a clinical evaluation of the expanded skin texture conducted by 2 experienced, independent senior plastic surgeons,[4] 35 patients were assigned to the well-regenerated group (13 in the control group, 11 in the adipose group, and 11 in the SVF group), whereas 31 were identified as having poorly regenerated skin (10 in the control group, 10 in the adipose group, and 11 in the SVF group). Patient demographics and baseline characteristics in the well-regenerated and the poorly regenerated subgroups are shown. (**See Table, Supplemental Digital Content 1**, which shows patient demographic and baseline characteristics in the well-regenerated and the poorly regenerated subgroups, ***https://links.lww.com/PRS/I214***.)

Intervention

After enrollment, for the adipose group, patients received an average volume of 30.26 ± 3.76 mL of fat injection. For the SVF group, a concentration of 1 × 10^6 cells/ml of SVF was injected, and the average number of cells injected was 39.86 ± 8.96 cells, with a cell proliferation rate of 89.45% ± 0.89%. The volume of fat and

Table 1. Patient Demographic and Baseline Characteristics

Characteristic	Control (%)	Control (%)	SVF (%)	P[b]
No.	23	21	22	
Age	30.9 ± 2.1[a]	28.2 ± 1.4[a]	27.8 ± 1.4[a]	0.373
Sex				
Male	10	11	11	—
Female	13	10	11	—
BMI	24.0 ± 0.9[a]	22.3 ± 0.8[a]	21.4 ± 0.8[a]	0.119
Expander implantation site				
Dorsum	3	3	4	—
Thoracoabdominal	7	9	10	—
Extremities	1	1	0	—
Facial and cervical	12	8	8	—
Designed expander volume				
≤200 mL	9	6	5	—
>200 to ≤400 mL	4	9	10	—
>400 to ≤600 mL	7	5	5	—
>600 mL	3	1	2	—
Expander implantation time, mo	6.6 ± 0.5[a]	8.4 ± 0.7[a]	7.1 ± 0.7[a]	0.160
EI	1.58 ± 0.14[a]	1.84 ± 0.17[a]	1.89 ± 0.20[a]	0.382
Subcutaneous fat donor area				
Periumbilical abdomen	—	9	8	—
Posterolateral thigh	—	12	14	—

BMI, body mass index.
[a]Data presented as mean ± SEM.
[b]$^{*}P<0.05$; $^{**}P<0.01$; $^{***}P<0.001$.

SVF injected varies according to the size of the expanded area in each patient.

Evaluation of Skin Thickness

At baseline, skin thickness across all groups was comparable, with the control group measuring 1.71 ± 0.08 mm, the adipose group measuring 1.63 ± 0.09 mm (–0.163 to 0.315; P = 0.736 versus control), and the SVF group measuring 1.52 ± 0.09 mm (–0.048 to 0.427; P = 0.138 versus control). However, by 12 weeks after treatment, both the adipose group (1.61 ± 0.07, –0.637 to –0.159; P < 0.001 versus control) and the SVF group (1.50 ± 0.06 mm, –0.526 to –0.054; P = 0.011 versus control) showed significantly greater thickness compared with the control group (1.21 ± 0.06 mm) (Fig. 2, *left*). (**See Table, Supplemental Digital Content 2**, which shows skin thickness and EI parameter at baseline [V1], and after 4 [V2], 8 [V3], and 12 [V4] weeks, ***https://links.lww.com/PRS/I215***.)

In the treated groups, skin thickness was largely maintained at baseline levels, whereas the control group showed marked thinning (Fig. 2, *right*), particularly in the dermis rather than the epidermis. (**See Figure, Supplemental Digital Content 3**, which shows comparison of skin thickness, including epidermis thickness and dermis thickness, across all groups, ***https://links.lww.com/PRS/I216***.)

Evaluation of Skin Thickness in Regeneration Categories

In evaluating skin thickness across different regeneration categories, distinct responses were noted between groups.

Well-Regenerated Subgroup

For cases with well-regenerated skin, both the SVF and adipose groups maintained significant thickness compared with the control group. By 12 weeks, the control group showed a marked decrease in skin thickness (1.35 ± 0.08 mm, 0.455 to 0.772; P < 0.001 versus baseline), whereas the SVF group (1.69 ± 0.08 mm, –0.280 to 0.515; P = 0.757 versus baseline) and the adipose group (1.71 ± 0.08 mm, –0.067 to 0.358; P = 0.203 versus baseline) preserved their skin thickness (Fig. 3). This improvement was primarily noted in the dermis, with both treatment groups showing significant enhancements (adipose, 1.41 ± 0.08 mm, –0.580 to –0.032; P = 0.027 versus control; SVF, 1.42 ± 0.08 mm, –0.580 to –0.050; P = 0.018 versus control) compared with the control group (1.11 ± 0.07 mm). (**See Table, Supplemental Digital Content 4**, which shows skin thickness and EI parameter in the well-regenerated subgroup at baseline [V1], and after 4 [V2], 8 [V3], and 12 [V4] weeks, ***https://links.lww.com/PRS/I217***. **See Figure, Supplemental Digital Content 5**, which

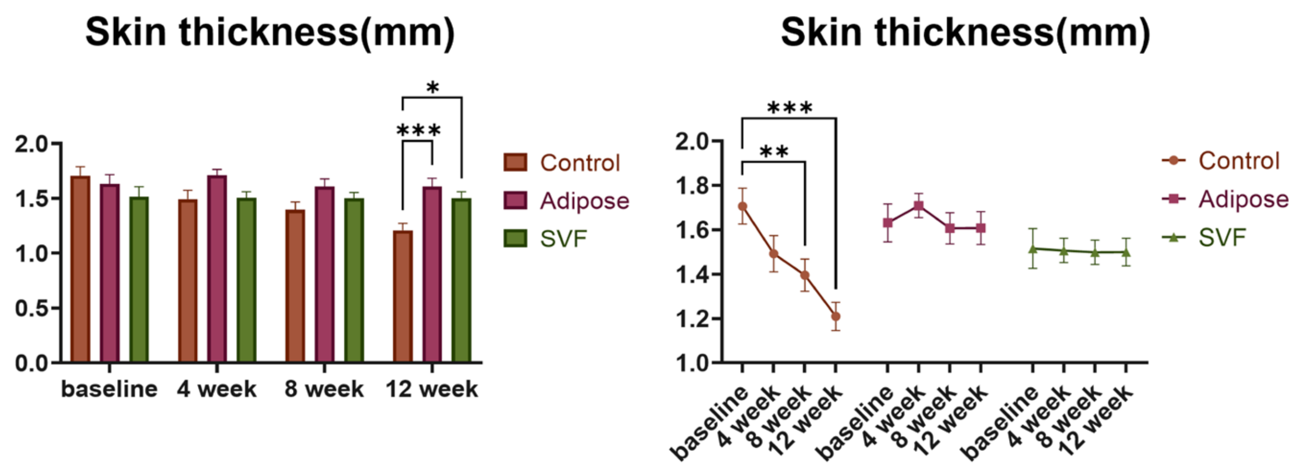

Fig. 2. Skin thickness comparison across treatment groups. Skin thickness comparison within each group from baseline to 12 weeks. Data indicate that both the adipose and SVF groups demonstrated significant increases in skin thickness at 12 weeks compared with the control group: $^{*}P < 0.05$; $^{**}P < 0.01$; and $^{***}P < 0.001$.

shows comparison of skin thickness, including dermal and epidermal thickness, in the well-regenerated subgroup, *https://links.lww.com/PRS/I218*.)

In the well-regenerated subgroup, a 24-year-old female control patient with a dorsal expander started with an EI of 2.33, showing mild telangiectasia in the expanded skin (Fig. 3). Over 12 weeks, as the EI increased to 3.57, the skin showed decreased thickness with stretch dermal striatum and telangiectasia. Another 24-year-old female patient in the adipose group, also with a dorsal expander, exhibited an initial EI of 1.59, characterized by prominent dermis striae and telangiectasia. After 12 weeks, the EI improved to 2.09, corresponding with increased skin thickness and a reduction in telangiectasia and dermal striae, suggesting enhanced tissue regeneration. In the SVF group, a 17-year-old female patient with a chest expander began with an EI of 3.18, displaying significant telangiectasia. After 12 weeks, the EI rose to 4.54, and despite ongoing telangiectasia, the EI was notably increased without new signs of deterioration.

Poorly Regenerated Subgroup

For patients with poorly regenerated skin, the SVF group (1.31 ± 0.04 mm, –0.691 to –0.131; P = 0.004 versus control) and the adipose group (1.50 ± 0.12 mm, –0.865 to –0.111; P = 0.011 versus control) also exhibited significant increases in skin thickness after 12 weeks compared with the control group (1.01 ± 0.06 mm). The control group continued to experience progressive thinning throughout the study (Fig. 4, *above*). The adipose group showed substantial dermal thickening by 4 weeks—a benefit that persisted throughout the 12-week period (control, 0.81 ± 0.07 mm; adipose, 1.18 ± 0.10 mm, –0.683 to –0.073; P = 0.014 versus control) (**see Figure, Supplemental Digital Content 5**, *https://links.lww.com/PRS/I218*). The SVF group contributed to both dermal (1.05 ± 0.06 mm, –0.467 to –0.013; P = 0.037 versus control) and epidermal (control, 0.22 ± 0.01 mm; SVF, 0.26 ± 0.01 mm, –0.072 to –0.003; P = 0.030 versus control) thickening, leading to a notable overall increase in skin thickness during the observation period. (**See Table, Supplemental Digital Content 6**, which shows skin thickness and EI parameter in the poorly regenerated subgroup at baseline [V1], and after 4 [V2], 8 [V3], and 12 [V4] weeks, *https://links.lww.com/PRS/I219*. **See Figure, Supplemental Digital Content 7**, which shows comparison of skin thickness, including dermal and epidermal thickness, in the poorly regenerated subgroup, *https://links.lww.com/PRS/I220*.)

In the poorly regenerated subgroup, a 30-year-old female control patient had a chest expander with an initial EI of 2.31, where the expanded skin had already thinned and showed telangiectasia with dermis striae (Fig. 4). After 12 weeks, the EI increased to 2.84, but the skin continued to deteriorate, with worsening dermal rupture, dermis striae and telangiectasia, leading to cessation of expansion. In the adipose group, a 44-year-old female patient with a chest expander had an initial EI of 3.38, presenting thin skin with telangiectasia and vascular

Fig. 3. (*Above*) Skin thickness comparison in well-regenerated group and the representative cases shown in *second row*, *third row*, and *below*. (*Second row*) Images from a 24-year-old woman in the control group illustrate progressive telangiectasia and stretch striae over a 12-week period. (*Third row*) A 24-year-old (*Continued*)

congestion. By 12 weeks, the EI increased to 4.52, which correlated with improved skin thickness and a significant reduction in telangiectasia, and dermis striae. In the SVF group, an 18-year-old female patient with a chest expander presented an initial EI of 1.58, with visible dermis striae and telangiectasia. Following 12 weeks, the EI reached 2.7, resulting in thicker skin and a marked reduction, in both the stretch striae and telangiectasia.

Overall, the expansion process typically resulted in skin thinning. However, interventions with SVF transplantation and fat grafting successfully countered this adverse outcome by significantly enhancing the dermal layers and, thus, overall skin thickness. Fat grafting was particularly more effective than SVF transplantation in achieving an increase in skin thickness.

Evaluation of EI

The EI demonstrated a marked increase from baseline to 12 weeks across all groups (Fig. 5). When compared with the control group (2.70 ± 0.23), both the adipose group (3.72 ± 0.21, –1.660 to –0.373; $P < 0.001$ versus control) and the SVF group (3.39 ± 0.16, –1.321 to –0.048; $P = 0.031$ versus control) effectively enhanced EI at 12 weeks (**see Table, Supplemental Digital Content 2**, ***https://links.lww.com/PRS/I215***). As a result, fat grafting and SVF transplantation increased the surface areas and volumes of the expanded tissues, suggesting that these treatments enabled a more effective and efficient expansion process.

Safety Assessment

Throughout the 12-week follow-up period, no participants experienced infectious or postinjection-related adverse events. Comprehensive monitoring over the 2-year follow-up further confirmed the absence of significant complications, with no instances of subcutaneous protrusion, mass formation, or hyperplasia at the injection sites.

DISCUSSION

Fat grafting extends beyond volume enhancement by secreting extracellular vesicles that aid in tissue repair, effectively addressing scarring and burns, and even autoimmune-related skin disorders.[19–21] Our study showed that fat grafting provided the greatest improvement in skin thickness, including both the epidermis and the dermis, benefiting both well-regenerated and poorly regenerated skin. Further analysis revealed that, in well-regenerated skin, this thickening effect was attributable to the injected fat helping the skin maintain its thickness without thinning. This may be because subcutaneously injected fat can provide structural support to reduce damaging mechanical forces on the skin. Excessive pressure can lead to skin thinning, necrosis, and delayed healing, such as pressure ulcers.[22] In poorly regenerated skin, fat transplantation directly increased skin thickness as early as 4 weeks after grafting. A study has shown that mechanical stretching prompts adipocytes to dedifferentiate into progenitor cells, which subsequently differentiate into fibroblasts and endothelial cells embedded within the dermis.[23] This process promotes cellular proliferation and increases vascularization, resulting in a significantly increasing dermal thickness.

SVF treatments work primarily through the secretion of cytokines and growth factors that stimulate the proliferation, migration, and differentiation of skin cells.[24–26] This mechanism leads to increased collagen deposition and angiogenesis in the wound, reduces inflammation, and fosters tissue healing outcomes—even under challenging conditions such as diabetic and ischemic wounds.[27,28] In this study, SVF treatment also demonstrated similar effects in thickening the skin and maintaining skin thickness. However, the effect of fat was superior to that of SVF, possibly because fat can provide multiple functions, including reducing excessive stretching pressure of the skin, and adipocytes can dedifferentiate into adipocyte progenitor cells to reconstruct the dermis and vasculature. Although both SVF and adipose tissue have the ability to secrete exosomes, adipose tissue secretes more exosomes with more diverse components, leading to rapid effect.[29]

By 12 weeks, the untreated expanded skin continued to thin, particularly in poorly regenerated areas, whereas fat grafting and SVF transplantation successfully countered this adverse outcome. These treatments have been shown to enhance vascularization, stimulate cell proliferation, and promote ECM synthesis, resulting in thickened skin and effectively rescuing poorly regenerating expanded skin.[11,12,15] However, these treatments differed depending on the state of regeneration. Well-regenerated groups maintained consistent

Fig. 3. (Continued). female patient in the adipose group demonstrating notable improvements in skin texture and an increase in skin thickness after treatment. (*Below*) A 17-year-old female patient from the SVF group with an improved EI and stable skin texture: $*P < 0.05$; $**P < 0.01$; and $***P < 0.001$.

Fig. 4. (*Above*) Skin thickness comparison in the poorly regenerated group and representative cases shown in the *second row* and the *third row*. (*Second row*) A 30-year-old female patient in the control group with worsening skin deterioration over 12 weeks. (*Third row*) Improvement in skin thickness and texture (*Continued*)

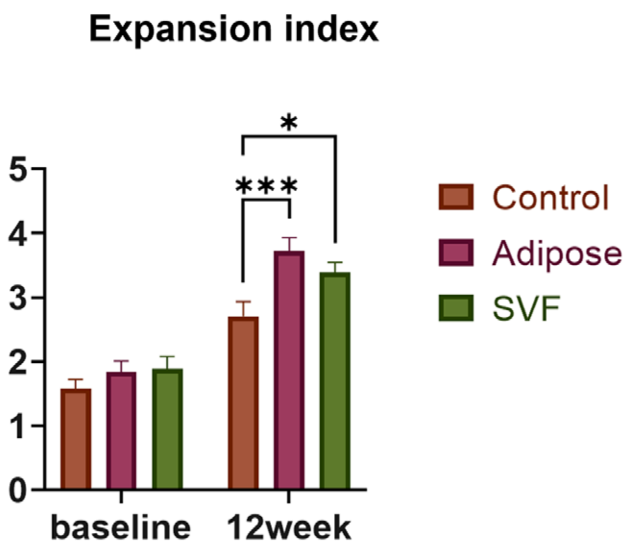

Fig. 5. EI changes. Both the adipose and SVF groups exhibited a significant increase in EI compared with the control group: $*P < 0.05$ and $***P < 0.001$.

skin thickness, whereas those with poorly regenerated skin showed substantial improvements following treatment, with increased skin thickness persisting from 4 to 12 weeks after treatment. These insights highlight the potential of adipose-derived treatments in treating areas with regenerative exhaustion.

Histologic evidence from our clinical trial demonstrated that SVF administration significantly enhanced angiogenesis, increased proliferating basal cells and fibroblasts, and promoted ECM remodeling.[15] These findings indicate that SVF stimulates tissue regeneration by means of cellular proliferation and structural reorganization. Our animal study showed that expanded skin with adipose tissue involvement had greater thickness, vascular density, and smooth muscle proliferation. Increased expression of paracrine factors—including vascular endothelial growth factor, epidermal growth factor, and basic fibroblast growth factor—in adipose tissue further supports its role in molecular crosstalk. Moreover, tracking adipose-derived cells confirmed their differentiation into fibroblasts and endothelial cells, directly contributing to skin regeneration.[24] These findings highlight that SVF-mediated and fat-assisted skin regeneration involves both cellular dynamics of basal cell and fibroblast proliferation and molecular signaling through paracrine effects of growth factors.

Overall, adipose-derived therapeutic strategies have significantly improved skin regeneration during expansion. Fat grafting and SVF treatment resulted in thicker skin that supported larger expansions, enabling the flap to cover extensive skin defects and improving reconstruction outcomes. (**See Figure, Supplemental Digital Content 8**, which shows fat grafting and SVF treatment improved reconstruction outcomes without postoperative complications. [*Above*] Images of a 30-year-old man in the adipose group. The patient underwent facial reconstruction for large facial scars using a cervical tissue expander. [*Below*] Images of a 17-year-old female patient in the SVF group. The patient underwent facial scar reconstruction using a facial tissue expander, ***https://links.lww.com/PRS/I221***.) These interventions effectively addressed issues such as insufficient skin and flap ischemia, thereby avoiding postoperative complications such as flap ischemia, poor incision healing, and secondary deformities. We have refined treatment protocols according to the stages of tissue regeneration, providing targeted strategies for both well-regenerated and poorly regenerated groups. However, the study has limitations, including the lack of a positive therapeutic control group, mainly because of the lack of effective alternative treatments for skin expansion. The use of saline injections following liposuction as a control group can ensure blinding of participants in the study; however, such controls may be inappropriate because of potential risks and ethical considerations. Future research should aim to expand the patient cohort within the treatment group and establish a control group to validate the robustness of the findings. Moreover, the technique of injecting treatments into expanded skin flaps carries a risk of expander rupture, highlighting the technical challenges of this approach.

Fig. 4. (Continued). reduction in telangiectasia for a 44-year-old female patient from the adipose group. (*Below*) A notable increase in skin thickness and decreased stretch striae and telangiectasia are noted in an 18-year-old female patient from the SVF group: $*P < 0.05$; $**P < 0.01$; and $***P < 0.001$.

CONCLUSIONS

Adipose-derived treatments, particularly SVF and fat grafting, represent a safe and effective strategy for enhancing skin regeneration during tissue expansion. These therapies effectively promote skin regeneration throughout the expansion process by maintaining skin thickness in well-regenerated skin and counteracting thinning in poorly regenerated skin. Our findings suggest that incorporating these treatments can optimize reconstructive procedures, and future research should focus on refining these therapeutic strategies and exploring their broader clinical applications.

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DISCLOSURE

The authors declare that there are no conflicts of interest regarding the publication of this article.

PATIENT CONSENT

Written informed consent was obtained from all individual participants included in this study.

ACKNOWLEDGMENTS

This study was supported by grants from the National Natural Science Foundation of China (82272287); the Shanghai Clinical Research Center of Plastic and Reconstructive Surgery, supported by the Science and Technology Commission of Shanghai Municipality (grant no. 22MC1940300); the Crossdisciplinary Research Fund of Shanghai Ninth People's Hospital, Shanghai Jiao Tong University School of Medicine (JYJC202215); the project list of "National Double First-Class" and "Shanghai-Top-Level" high education initiative at Shanghai Jiao Tong University School of Medicine; and Shanghai Key Research Center–Shanghai Research Center for Plastic Surgery (2023ZZ02023). The authors gratefully acknowledge the contribution to this work by Drs. Kai Liu, Feng Xie, Bin Gu, Tao Zan, Haizhou Li, Yashan Gao, Ru-lin Huang, Dong Han, and Jizhou He by referring their patients to this study.

AVAILABILITY OF DATA AND MATERIAL

The datasets used or analyzed for the current study are available from the corresponding author upon reasonable request.

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